



// IDP AS A SERVICE: MANAGING TENANTS EFFICIENTLY USING GITOPS

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Slides



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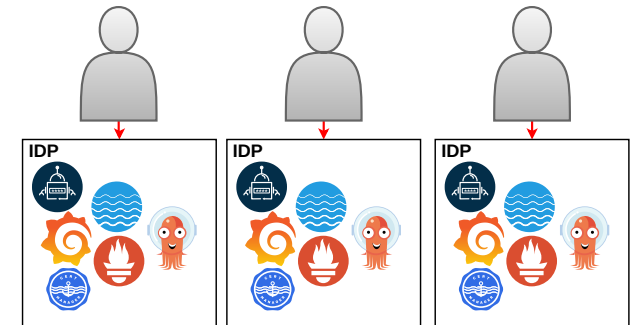
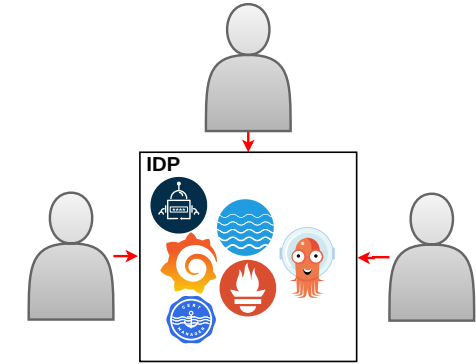
Agenda

- Shared IDP Instance
- Dedicated IDP Instances (IDPaaS)
- Multiple Environments per Tenant

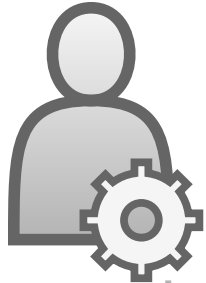


Multi-tenancy strategies

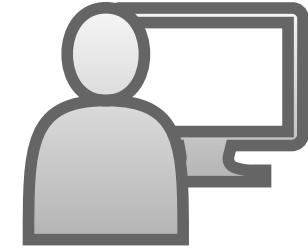
- Shared instance
- Dedicated instances



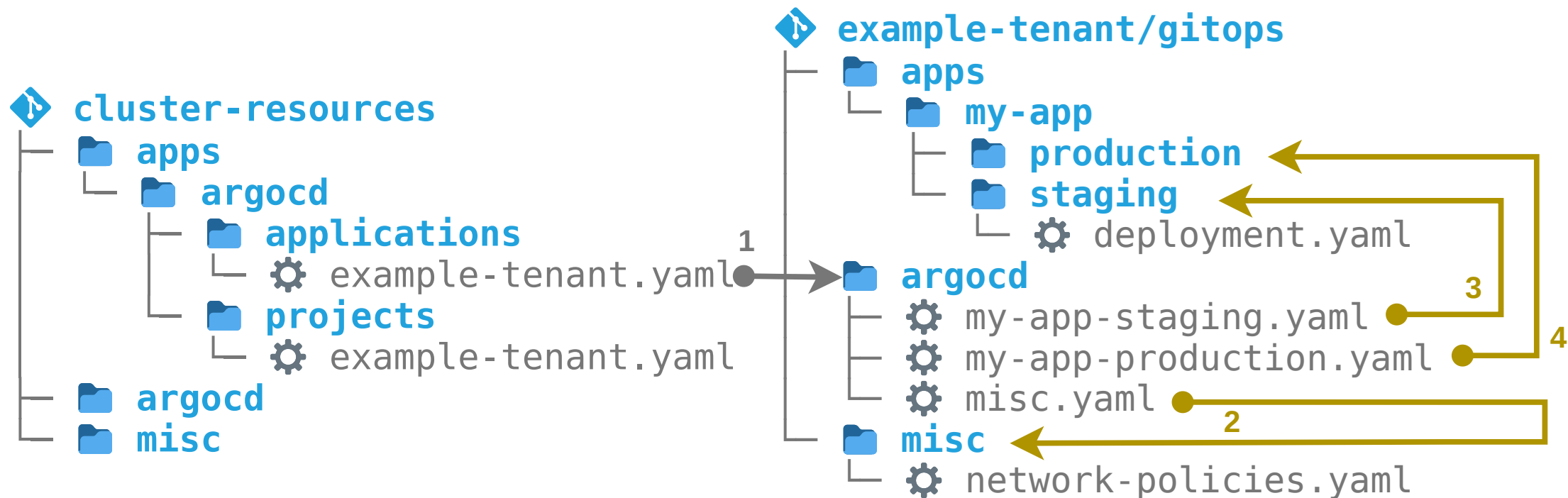
Shared Instance: A first approach



Platform engineer



Software engineer



 [cloudogu/gop-multi-tenant-shared-example](https://github.com/cloudogu/gop-multi-tenant-shared-example)

Don't forget to authz



```
policy.csv: |
# groups: Mapping from groups to RBAC roles
g, tenant1, role:role-tenant1
# policies:
g, tenant1, role:role-tenant1
p, role:role-tenant1, applications, get, role-tenant1/*, allow
p, role:role-tenant1, applications, sync, role-tenant1/*, allow
p, role:role-tenant1, applications, get, argo-project/role-tenant1-control-app, allow
p, role:role-tenant1, applications, sync, argo-project/role-tenant1-control-app, allow
p, role:role-tenant1, projects, get, role-tenant1, allow
p, role:role-tenant1, clusters, get, "https://kubernetes.default.svc", allow
p, role:role-tenant1, repositories, get, "ssh://...tenant1/gitops", allow
```



```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  # Allows users read-only access to all resources
  name: viewers
  namespace: tenant1
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: view
subjects:
- apiGroup: rbac.authorization.k8s.io
  kind: Group
  name: tenant1
```

 github.com/argoproj/argo-cd/blob/v3.2.0/docs/operator-manual/rbac.md

Shared vs Dedicated Instance

- Lower infra usage
- Easier to maintain
- Higher isolation
- Easier to configure

 blog.argoproj.io/best-practices-for-multi-tenancy-in-argo-cd-273e25a047b0

Dedicated instances: Options

Topology:

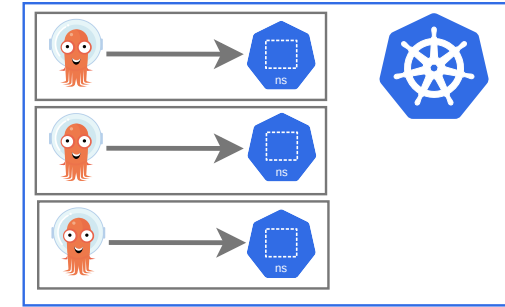
- Instance per Namespace
- Instance per Cluster

Management:

- Standalone (decentralized, autonomous)
- Hub and Spoke (centralized, platform)

IDP as a Service

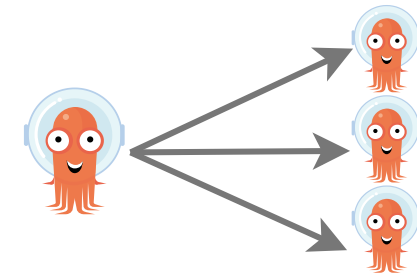
💡 Argo CD AppSets



Instance per Namespace

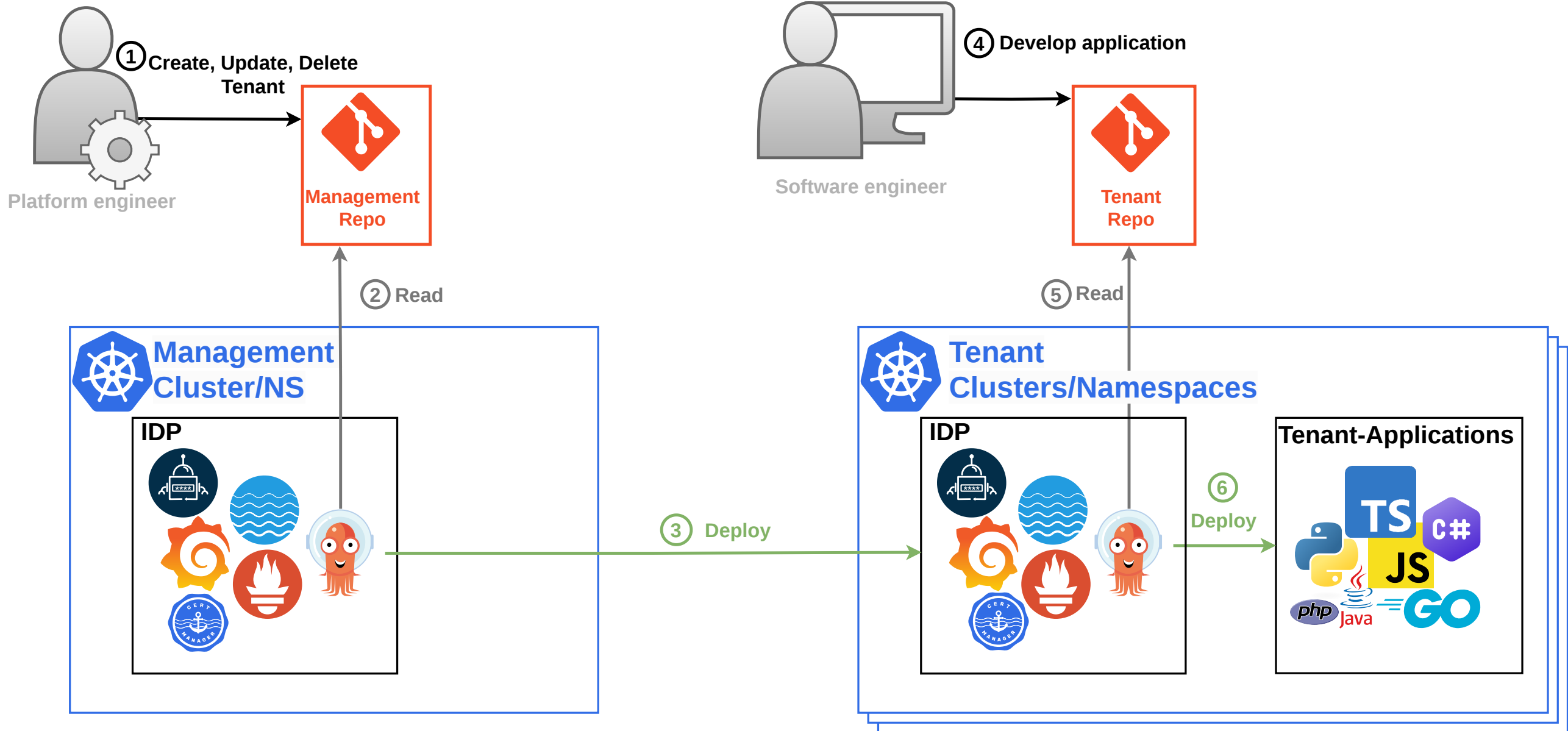


Instance per Cluster

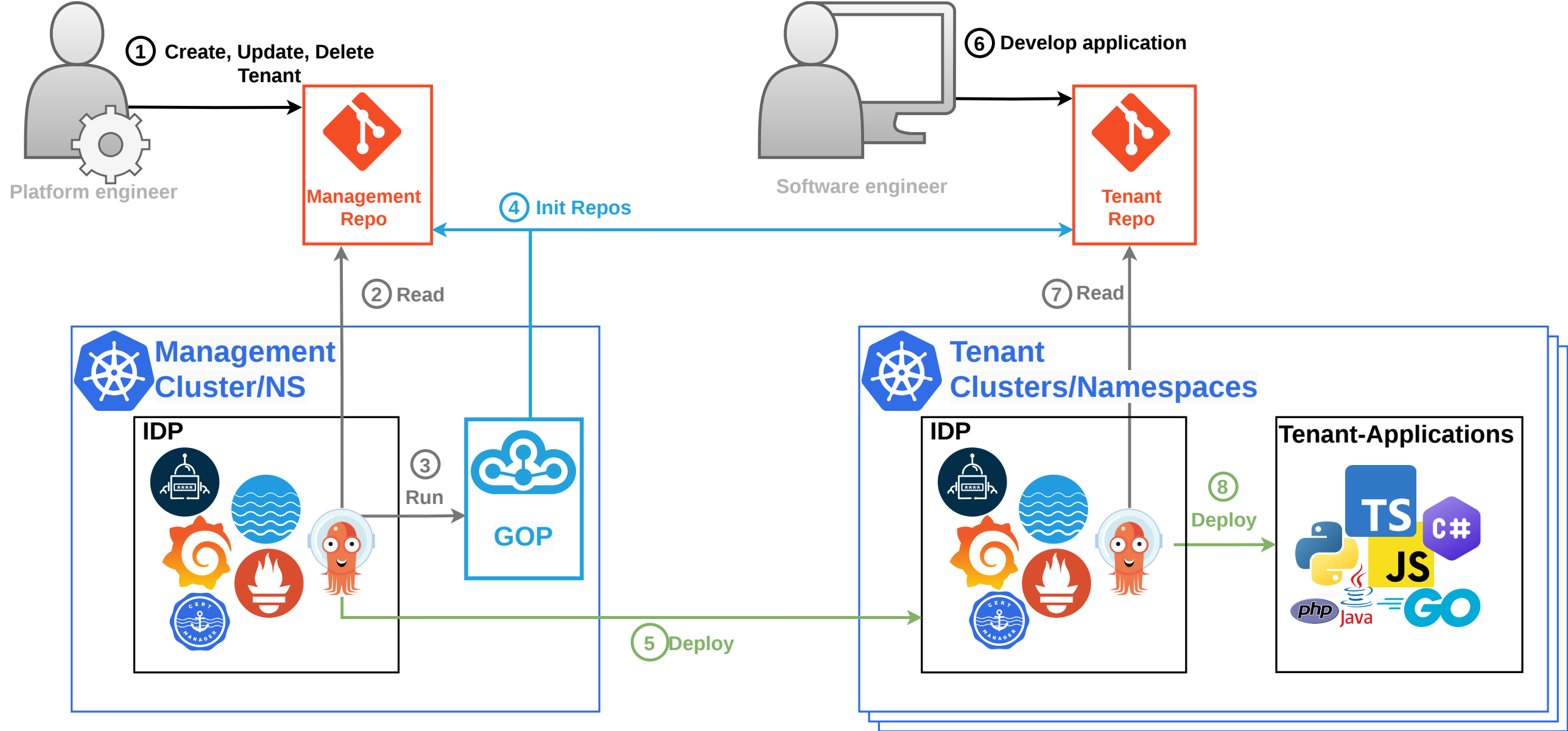


Hub and Spoke

IDP as a Service using Argo CD AppSets



IDP as a Service using Argo CD AppSets





1

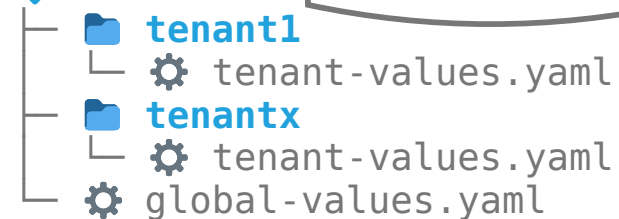


2

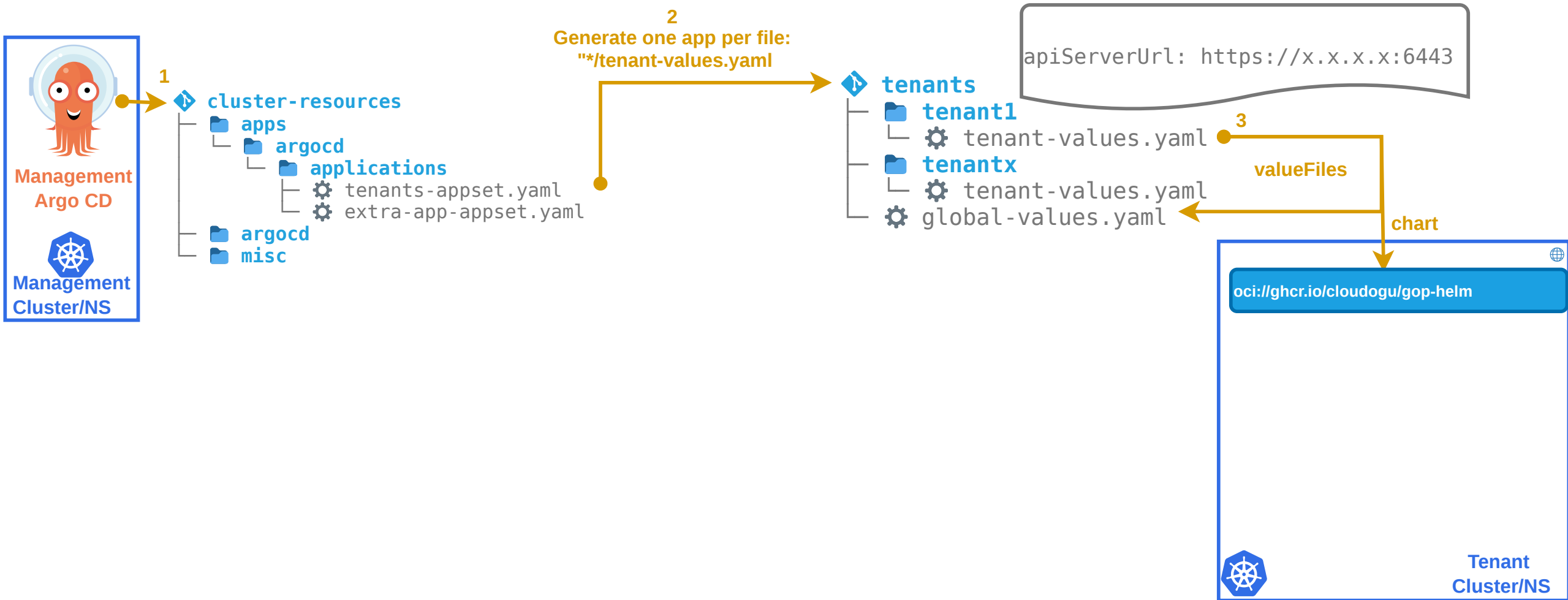
Generate one app per file:
"/tenant-values.yaml"

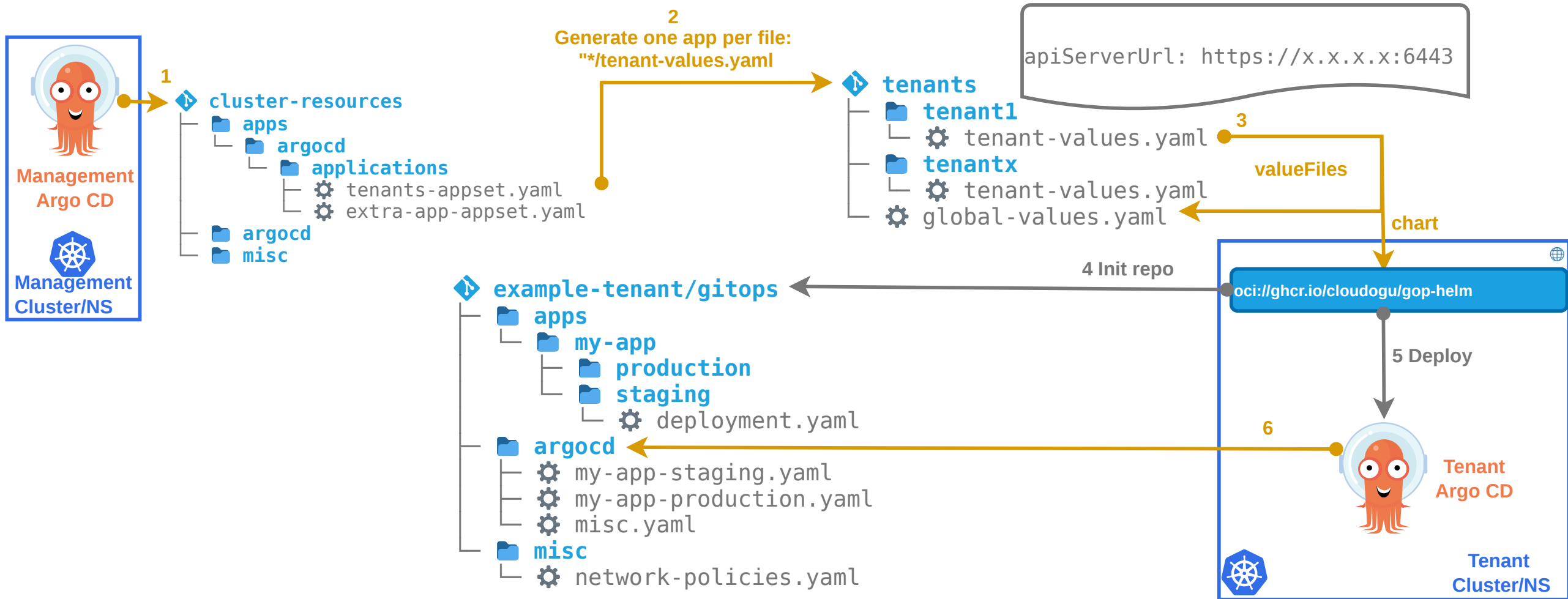


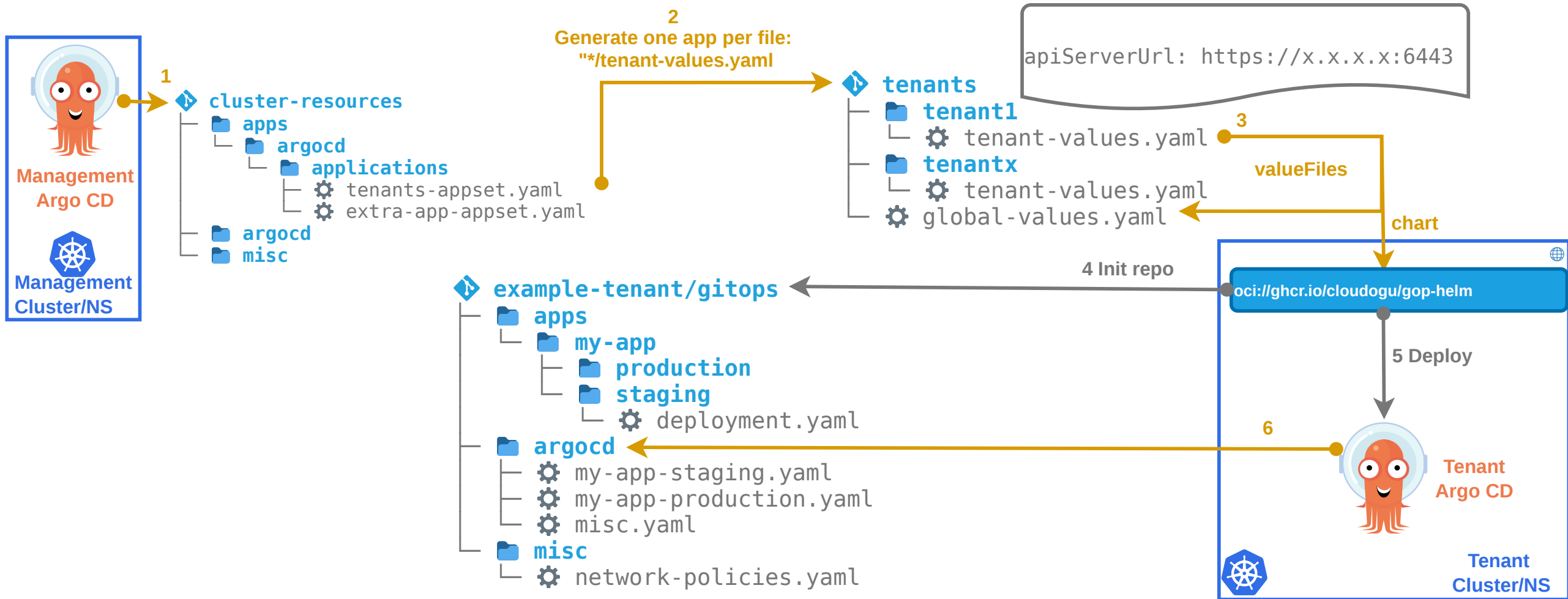
tenants



apiServerUrl: https://x.x.x.x:6443







 [cloudogu/gop-multi-tenant-multi-cluster-example](https://github.com/cloudogu/gop-multi-tenant-multi-cluster-example)

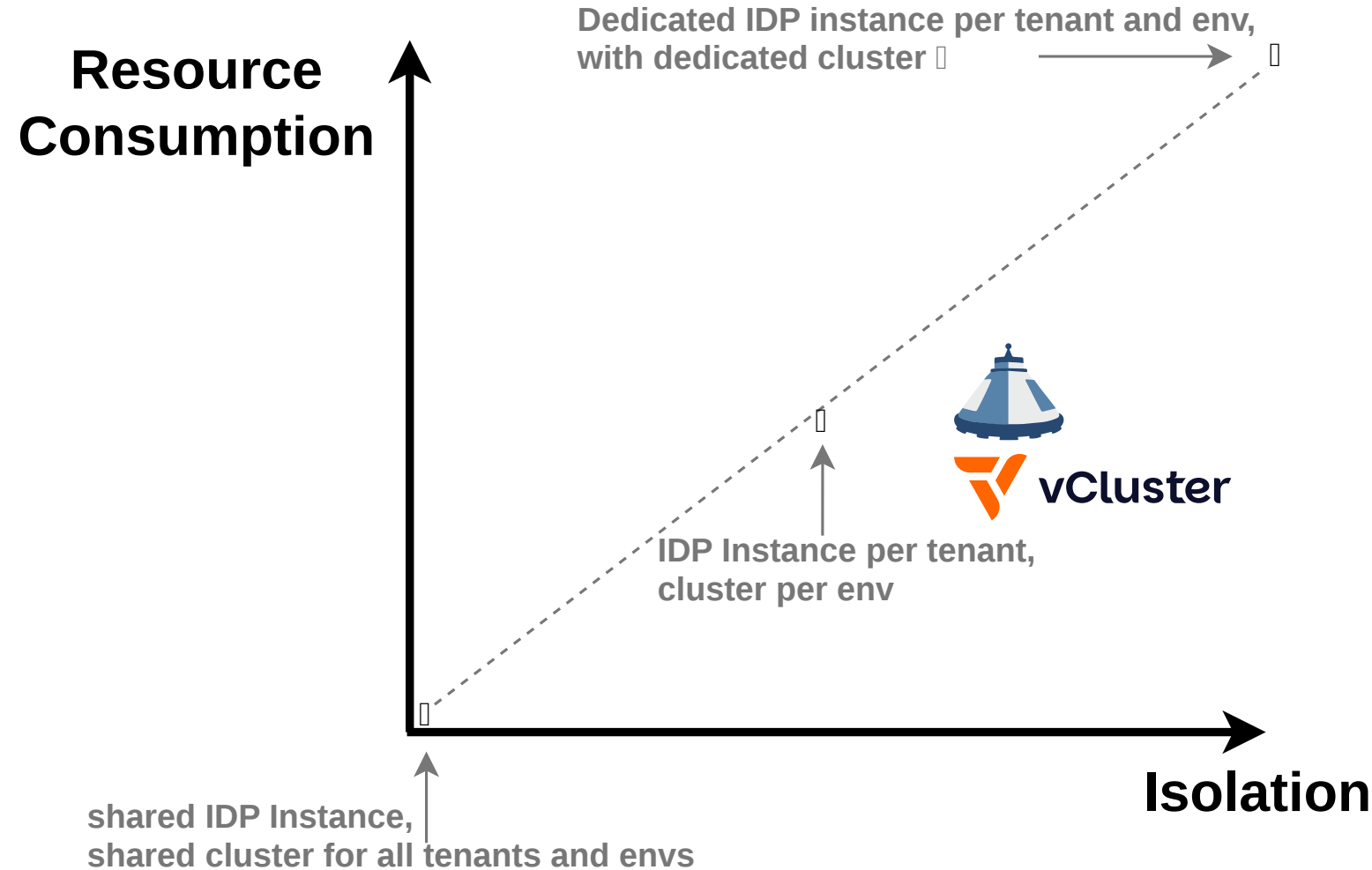
Instance per Namespace vs per Cluster

- Lower infra usage
- Easier to maintain
- Higher isolation
- Easier to configure    ...

 github.com/argoproj-labs/argocd-operator

What about env/stages?

Each tenant might have multiple envs

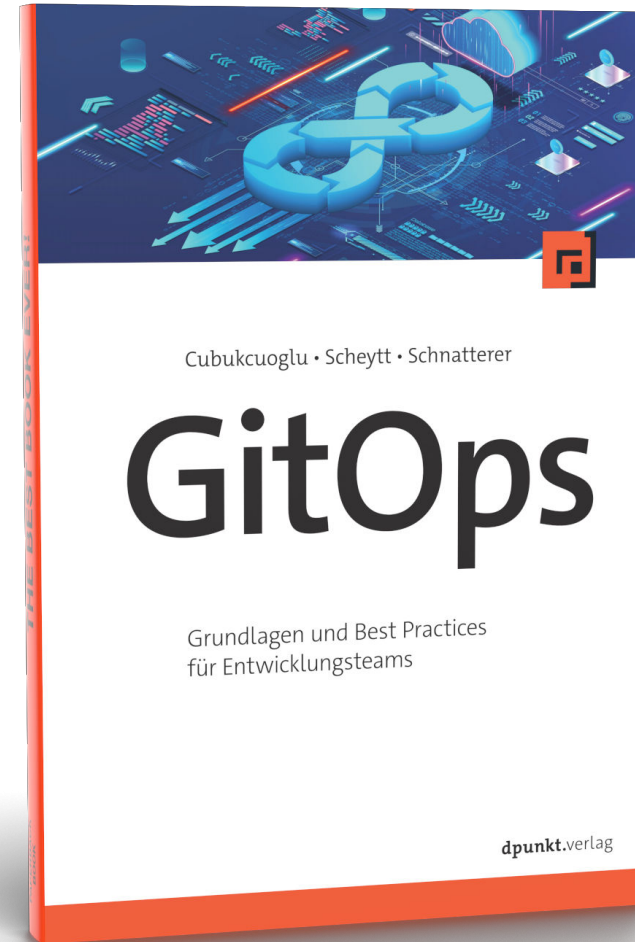
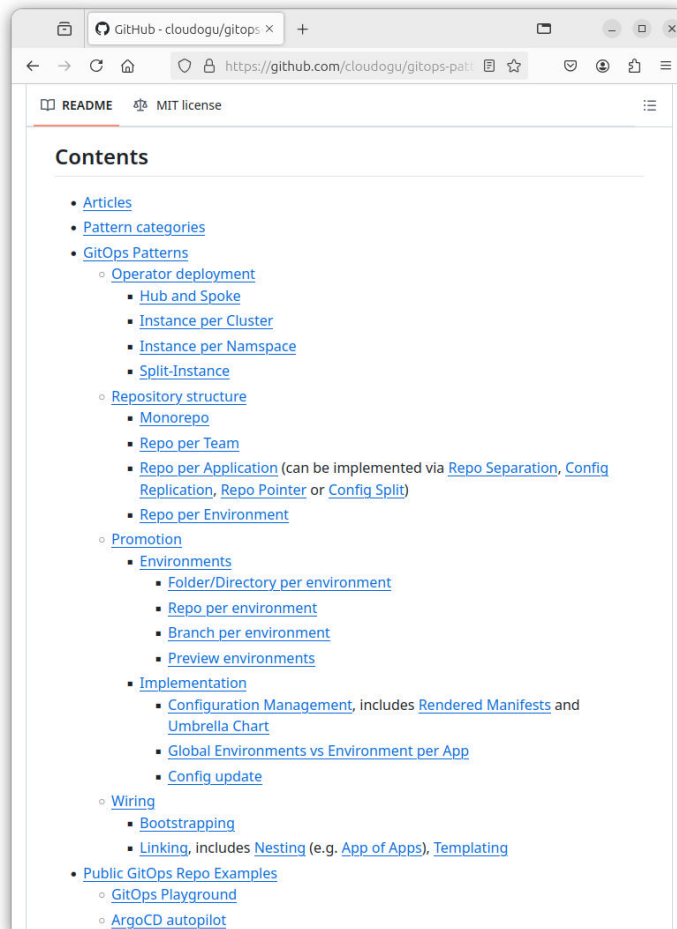


My key experiences: IDPs and tenants

- **Shared Instance:** 👍 for small settings. Lots of authz in tools.
- **Dedicated instances:** 🔒 Better isolation+UX but more effort
 - per Namespace: Less infra, but lots of RBAC, NetPols, etc.
 - per Cluster: More infra+complexity, but authz easier.
 - IDPaaS using AppSets 🐙 great to have but hard to get.
- No silver bullet: Number of tenants/namespaces/clusters depends on requirements,
e.g. envs, security, regulation, onPrem/public cloud, etc.



Further reading



IDP Bootstrapping made simple: GOP

Creates a complete GitOps-based operational stack / IDP on your Kubernetes clusters

 [cloudogu/gitops-playground](https://github.com/cloudogu/gitops-playground)

Also runs locally



```
VERSION='0.12.1'
bash <(curl -s \
  "https://raw.githubusercontent.com/cloudogu/gitops-playground/$VERSION/scripts/init-cluster.sh") \
  && docker run --rm -t -u $(id -u) \
    -v ~/.config/k3d/kubeconfig-gitops-playground.yaml:/home/.kube/config \
    --net=host \
    ghcr.io/cloudogu/gitops-playground:$VERSION --yes --argocd --ingress-nginx --base-url=http://localhost
# More IDP-features: --monitoring --vault=dev --cert-manager --mailhog
# More features for developers: --jenkins --registry --content-examples
```

-  [cloudogu/gop-multi-tenant-shared-example](https://github.com/cloudogu/gop-multi-tenant-shared-example)

Try with GOP: •  [cloudogu/gop-multi-tenant-multi-cluster-example](https://github.com/cloudogu/gop-multi-tenant-multi-cluster-example)

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Slides



☒ Let's meet



Image sources

- Live Example:
<https://unsplash.com/photos/assorted-color-hot-air-balloons-during-daytime-DuBNA1QMpPA>
- More examples:
<https://unsplash.com/photos/XZc4f2XZc84>
- coloured-parchment-paper background by brgfx on Freepik:
https://www.freepik.com/free-vector/coloured-parchment-paper-designs_1078492.htm